

Python Basics

Python was invented by Guido van Rossum.

Python is a high-level, interpreted programming language. It is easy to learn and widely used in AI, Data Science, Automation, and Web Development.

Guido van Rossum was born in 1978

Variables

Variables store values.

Example: `x = 10`

`name = 'Sushma'`

Data Types

Common data types: int, float, str, bool, list, tuple, set, dict.

Functions

Functions are reusable blocks of code.

Example:

```
def add(a,b):
```

```
    return a+b
```

Object Oriented Programming

Important concepts: Class, Object, Inheritance, Polymorphism, Encapsulation, Abstraction.

NumPy

Used for fast numerical computing with arrays.

Pandas

Used for data analysis using DataFrames.

Matplotlib

Used for data visualization.

Machine Learning

Machine Learning enables computers to learn patterns from data.

Deep Learning

Deep Learning uses neural networks with multiple layers.

Natural Language Processing

NLP helps computers understand and generate human language.

Generative AI

Generative AI creates text, images, audio, or code using foundation models.

RAG

Retrieval-Augmented Generation retrieves relevant documents before generating an answer.

Embeddings

Embeddings convert text into numerical vectors that capture semantic meaning.

Vector Databases

Examples: ChromaDB, FAISS, Pinecone, Weaviate.

Prompt Engineering

The process of writing effective prompts to improve LLM responses.

LangChain

Framework for building LLM and RAG applications.

Groq

Cloud platform providing very fast inference for open-source LLMs.

Interview Questions

What is RAG? What are embeddings? Why chunking? Difference between FAISS and ChromaDB?